

Fall 2025 Past Quizterm 1 Questions

Question 1

shape: (3, 2)

a	b
i64	i64
5	8
3	7
6	4

The DataFrame above is called `df`, and `polars` has been imported as `pl`. Consider the following code snippets. If the code snippet will return an integer between 0 and 8, **write the integer that is returned**. If the code snippet will return something other than an integer between 0 and 8, or will return an error, **write NA**.

Note: This question has been translated to Polars from Pandas. Keep in mind that there could be translation errors. Let us know if you spot any!

(i) [1.5 Pt] `df['b'][1]`

i 7 ?

(ii) [1.5 Pt] `df['b', :][1]`

ii NA ?

(iii) [1.5 Pt] `df[1, 2]`

iii NA ?

(iv) [1.5 Pt] `df.filter(pl.col('b') == 4)['a'][0]`

iv 6 ?

(v) [1.5 Pt] `df.sort('a', descending=True).head(1)['b'][0]`

v 4 ?

Question 2

The `DataFrame` below is called `df`, and `polars` has been imported as `pl`. Consider the following code snippets. Each snippet returns a `DataFrame`. **Enter the number of rows** in the returned `DataFrame`. If there is not enough information provided to answer the question, **enter NA**.

shape: (7, 2)

a	b
<i>str</i>	<i>i64</i>
"bye"	5
"bye"	9
"hello"	6
"hello"	7
"hello"	8
"hi"	3
"hi"	4

Note: This question has been translated to Polars from Pandas. Keep in mind that there could be translation errors. Let us know if you spot any!

(a) [1.5 Pt] `df.group_by('a').agg(pl.col('b').min())`

a ?

(b) [1.5 Pt] `df.group_by('a').head(2)` *2+2+2*

b ?

(c) [1.5 Pt] `df.filter(pl.col('b').min().over('a') < 8)` *every group min < 8, all rows pass*

c ?

Question 3

Consider the following sentence:

"Ann and Bob have aged 51 years since the year 1974."

For each of the regular expressions in the table below, write the matching substring(s) from the provided sentence. Make sure to write substring matches in the order they appear in the text.

- The first matching substring should go under **Match 1**, the second matching substring under **Match 2**, and the third matching substring under **Match 3**.
- If there are no matches, write **NA** in all of the boxes. If there is only one match, write **NA** in the Match 2 and Match 3 boxes. If there are only two matches, write **NA** in the Match 3 box.

For your reference, the sentence is shown again below. We have also included an example table illustrating the format of the answer table.

"Ann and Bob have aged 51 years since the year 1974."

	regex	Match 1	Match 2	Match 3
0	be.n	been	NA	NA
1	h.	ha	he	NA
2	\d+	(a)	(b)	(c)
3	[A-Z][a-z]	(d)	(e)	(f)
4	[aeiou]{2}	(g)	(h)	(i)

Click the DataFrame to open the image in a new tab.

regex	Match 1			Match 2			Match 3		
\d+	a	51	?	b	1974	?	c	NA	?
[A-Z][a-z]	d	An	?	e	Bo	?	f	NA	?
[aeiou]{2}	g	ea	?	h	ea	?	i	NA	?

Question 4

Rohan would like to determine whether the gym is more crowded on hotter or colder days. For several days, Rohan asks students at the gym to record their total workout time (in minutes). He also records the temperature outside during their workout (in °F).

Here are the first five rows of the data Rohan collected. There are 1000 rows in the full `DataFrame`:

	temperature	workout_time
0	63.1	31.3
1	83.3	49.7
2	75.6	38.8
3	71	35.2
4	55.5	22.7

(a) [4 Pts] Rohan wants to visualize the distribution of `workout_time`. Mark **True** if the plot type is appropriate for this visualization task, and **False** otherwise.

Histogram ☒ True ☐ False

Scatter plot ☐ True ☒ False

KDE Plot ☒ True ☐ False

Boxplot ☒ True ☐ False

(b) [4 Pts] Rohan wants to visualize the correlation of `temperature` and `workout_time`. Mark **True** if the plot type is appropriate for this visualization task, and **False** otherwise.

Note: Assume that the data type cannot be changed before plotting. For example, `temperature` must be plotted as a quantitative variable. It cannot be converted into qualitative ordinal variable.

Hexplot ☒ True ☐ False

Overlaid histograms ☐ True ☒ False

Contourplot ☒ True ☐ False

Side-by-side Boxplots ☐ True ☒ False

Question 5

Josh wants to learn whether UC Berkeley students prefer eating cake or pie. With help from the registrar, he collects a simple random sample (SRS) of **100** emails from the official list of all UC Berkeley students. He emails each selected student and asks them to report whether they prefer cake or pie. **62** students respond to his email within one week. Josh received no additional responses after that.

Jake tells Josh that he could have improved his survey by making two updates:

- Josh should email an SRS of 200 UC Berkeley student emails, instead of 100.
- Josh should also send a **reminder** email to all students who did not open the initial email within one week.

Assume that Josh could go back in time and make both updates suggested by Jake, and no other changes. Answer the following questions:

Note: You can assume that everyone who fills out either survey will always answer truthfully.

(i) [1 Pt] The non-response bias of the **updated** survey is likely _____ the non-response bias of the **original** survey.

- ☐ (a) Larger than
- ☒ (b) Smaller than
- ☐ (c) The same as

(ii) [1 Pt] The selection bias of the **updated** survey is likely _____ the selection bias of the **original** survey.

- ☐ (a) Larger than
- ☐ (b) Smaller than
- ☒ (c) The same as

(iii) [1 Pt] The chance error of the **updated** survey is _____ the chance error of the **original** survey.

- ☐ (a) Larger than
- ☒ (b) Smaller than
- ☐ (c) The same as

(iv) [1 Pt] The response bias of the **updated** survey is likely _____ the response bias of the **original** survey.

- ☐ (a) Larger than
- ☐ (b) Smaller than
- ☒ (c) The same as

Question 6

a) [2 Pts] For a constant model fit to a dataset with at least one datapoint, there is _____ exactly one parameter value that minimizes the **MAE**.

- ☐ (a) Always
- ☒ (b) Sometimes, but not always
- ☐ (c) Never

b) [2 Pts] For a given constant model and dataset, minimizing the **sum** of loss across all data points will _____ result in the same values of the optimal parameter(s), compared to minimizing the **average** loss across all datapoints.

- ☒ (a) Always
- ☐ (b) Never
- ☐ (c) Sometimes, but not always

	x	$\hat{y}$	y
0	2	4	3
1	-3	0	1
2	7	-1	0

c) [2 Pts] The three datapoints and predictions above are from a fitted SLR model. Compute the **MSE** of the SLR model. Your final answer should be an integer.

MSE integer value

|

[Optional] To be eligible for partial credit, you can optionally show your work in the box below:

Work $residual = (-1, 1, 1) \quad \frac{(-1)^2 + (1)^2 + (1)^2}{3} = 1$

Question 7

The partial derivative of the objective function $\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2)^2$ with respect to θ_2 is:

- ☐ (a) $\frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0 - \theta_1 x_i - 2\theta_2 x_i)$ $\frac{1}{n} \sum 2 (\downarrow) (-x_i^2)$
- ☐ (b) $\frac{1}{n} \sum_{i=1}^n 2(-\theta_1 - 2\theta_2 x_i)$
- ☐ (c) $\frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2)(-\theta_1 x_i - 2\theta_2 x_i)$
- ☐ (d) $-\frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2)$
- ☒ (e) None of the above

Spring 2026 Past Quizterm 1 Questions

Question 1

	I	A	M	B
0	21	18	23	13
1	19	10	25	11
2	20	15	16	12
3	22	24	17	14

4 rows x 4 columns

The DataFrame above is called `df`.

(a) [2 Pts] For each line of code below, write the integer it returns (e.g., `10`). If it returns something other than an integer or produces an error, write `NA`.

`df.sort('B')['I'][0]`

a ?

`df['I'][0]`

b ?

`df[1, 'B']`

c ?

`df.select('B')[1]` Return 1-row df

d ?

(b) [1.5 Pts] For each line of code, write the **number of rows** in the DataFrame it returns (e.g., `2`). If the resulting DataFrame has no rows, write `0`. If the expression does not return a DataFrame or returns an error, write `NA`.

`df['M'] == 25` Return Series

a ?

`df.filter((pl.col('A') > 23) | (pl.col('M') < 24))`

b ?

`df.filter((pl.col('I') > 19) & (pl.col('A') <= 23))`

c ?

Question 2

a	b
chow mein	11
chow mein	20
linguine	4
spaghetti	21
spaghetti	14
linguine	8
spaghetti	11

The `DataFrame` above is called `df`, and `polars` has been imported as `pl`. Each subpart will show you a line of code involving `df` and then a copy of `df` for your convenience. Each line returns a `DataFrame` or an error. For each line, you should choose the correct `DataFrame` or error that would be returned.

Note: This question has been translated to Polars from Pandas. Keep in mind that there could be translation errors. Let us know if you spot any!

(a) [1.5 Pt] Select the correct `DataFrame` that would result from the code.

```
df.group_by('a').agg(pl.col('b').min()).sort('a')
```

- ☐ (a)

a	b
chow mein	4
linguine	4
spaghetti	4

☐ (b)

a	b
chow mein	4

☐ (c)

b
4

☐ (d)

a	b
linguine	4

☒ (e)

a	b
chow mein	11
linguine	4
spaghetti	11

☐ (f) None of these [Clear](#)

(b) [1.5 Pt] Select the correct `DataFrame` that would result from the code.

```
df.group_by('b').head(1).sort('b')
```

- ☐ (a) The code will return an error.
☐ (b)

b
11

☐ (c)

a	b
chow mein	11
linguine	4
spaghetti	21

☒ (d)

b	a
4	linguine
8	linguine
11	chow mein
14	spaghetti
20	chow mein
21	spaghetti

☐ (e)

b	a
4	linguine
8	linguine
11	chow mein
11	spaghetti
14	spaghetti
20	chow mein
21	spaghetti

☐ (f)

a	b
linguine	4
linguine	8
chow mein	11
spaghetti	14
chow mein	20
spaghetti	21

[Clear](#)

(c) [1.5 Pt] Select the correct `DataFrame` that would result from the code.

```
df.filter(pl.col('b').max().over('a') < 13)
```

- ☐ (a)

a	b
chow mein	20
spaghetti	21
spaghetti	14

☐ (b)

a	b
linguine	8

☐ (c)

a	b
chow mein	11
linguine	4
linguine	8
spaghetti	11

☐ (d)

a
chow mein
spaghetti

☒ (e)

a	b
linguine	4
linguine	8

☐ (f)

a
linguine

[Clear](#)

Question 3

Consider the following sentence:

Twenty years ago, during 05/2050, aliens seized TWO worlds.

For each of the regular expressions in the table below, write the matching substring(s) from the provided sentence. Make sure to write substring matches in the order they appear in the text.

- The first matching substring should go under **Match 1**, the second under **Match 2**, and the third under **Match 3**.
- If there are **no matches**, write **NA** in all boxes for that row. If there is only one match, write **NA** in the Match 2 and Match 3 boxes. If there are only two matches, write **NA** in the Match 3 box.
- If there are more than 3 matches, write only the first 3 in order.

Below is an example showing the expected format:

regex	Match 1	Match 2	Match 3
rs\sag	rs ag	NA	NA
\d{2}[^0-9]	(a)	(b)	(c)
^[Ww]?	(d)	(e)	(f)
\w+\.	(g)	(h)	(i)

Here's the sentence again for your reference:

Twenty years ago, during 05/2050, aliens seized TWO worlds.

regex	Match 1			Match 2			Match 3		
\d{2}[^0-9]	a	05	?	b	50	?	c	NA	?
start ^[Ww]?	d	Tw	?	e	NA	?	f	NA	?
\w+\.	g	worlds.	?	h	NA	?	i	NA	?

Question 4

(a) Histogram and KDE [1.5 Pts]



Consider a dataset with two data points: **10** and **20**. Consider a histogram of this dataset using two bins. The first bin has a lower bound at **5** and an upper bound at **15**. The second bin has a lower bound at **15** and an upper bound at **25**.

What is the height of each bar in the histogram? If your answer is a decimal, do NOT round your answer. You may also write your answer as a fraction.

Note: All histograms in Data 100 have density on the y-axis, not count or proportion.

Height of first bar = ?

Height of second bar = ?

Suppose we make a KDE curve of the same dataset using the boxcar kernel, which is defined below. What value of α will make this KDE curve identical to the histogram above? If no such value exists, write **NA**.

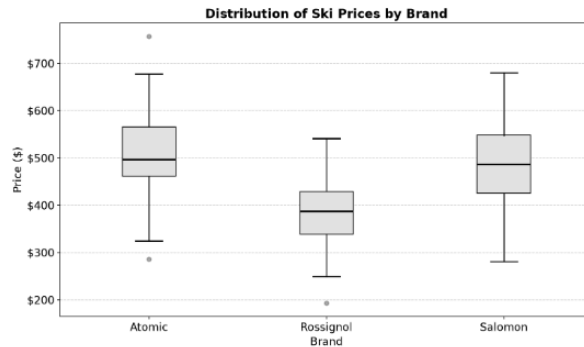
α = ?



This is a boxcar kernel centered at 0 with a bandwidth parameter of 1 (i.e., $\alpha=1$).

(b) Skis [2.25 Pts]

The box plot below was generated using a **single call** to `sns.boxplot()` on a pandas DataFrame called `skis`. Note: Assume that the `skis` dataframe was passed directly to `sns.boxplot()` with no modifications and that no additional functions are used inside of `sns.boxplot()`.



Question

[1 Pts] What is the **minimum number of columns** `skis` needs to have in order to generate this plot using `sns.boxplot()`?

Your Answer

integer ?

[1.25 Pts] Which of the following plot types is an appropriate choice for visualizing the same column(s) of data used by the side-by-side boxplots? Select **True** if the plot can visualize the same data, and **False** if not. Assume that you cannot make any transformations or aggregations of the underlying data and you cannot use additional columns or data sources.

Visualization

True / False

Overlaid histograms

☒ True ☐ False [Clear](#)

Overlaid KDEs

☒ True ☐ False [Clear](#)

Contour Plot

☐ True ☒ False [Clear](#)

Hex Plot

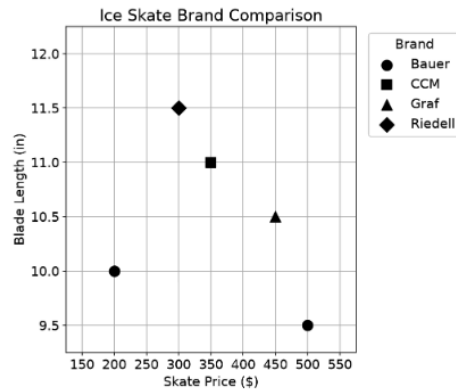
☐ True ☒ False [Clear](#)

Lineplot

☐ True ☒ False [Clear](#)

(c) Skates [2 Pts]

The scatter plot below was generated using a **single call** to `sns.scatterplot()` on a pandas DataFrame called `skates`. Note: Assume that the `skates` dataframe was passed directly to `sns.scatterplot` with no modifications and that no additional functions are used inside of `sns.scatterplot`.



You should assume there is no overplotting.

Question

How many **rows** does `skates` need to have in order to generate this plot?

What is the **minimum number of columns** `skates` needs to have in order to generate this plot using `sns.scatterplot()`?

Your Answer

integer

integer

Question 5

Consider this list of 8 objects:

X0XAX0AX

For each sample below (written as a length-3 or length-4 string of symbols), select **Possible to observe** if it's a possible sample according to the instructions above it, or **Impossible to observe** if it's not.

a) [0.3 Pts] Sample 3 objects uniformly at random from the list *with replacement*

XXX ☒ Possible to observe ☐ Impossible to observe [Clear](#)

X0A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

000 ☒ Possible to observe ☐ Impossible to observe [Clear](#)

b) [0.3 Pts] Sample 3 objects uniformly at random from the list *without replacement*

XXX ☒ Possible to observe ☐ Impossible to observe [Clear](#)

X0A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

000 ☐ Possible to observe ☒ Impossible to observe [Clear](#)

c) [0.3 Pts] A simple random sample (SRS) of size 3

XXX ☒ Possible to observe ☐ Impossible to observe [Clear](#)

X0A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

000 ☐ Possible to observe ☒ Impossible to observe [Clear](#)

d) [0.3 Pts] One bootstrap sample (size 8 with replacement)

XXXXXXXX ☒ Possible to observe ☐ Impossible to observe [Clear](#)

XXXX00AA ☒ Possible to observe ☐ Impossible to observe [Clear](#)

AAAAAA ☒ Possible to observe ☐ Impossible to observe [Clear](#)

e) [0.3 Pts] A stratified random sample of size 4, stratified by symbol.

There are three different symbols in the list of objects.

XXXX ☐ Possible to observe ☒ Impossible to observe [Clear](#)

XX0A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

X00A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

Question 6

Suppose we have a mixed population of only two types of flying bugs: **honeybees** and **hornets**. You gather a small sample of bugs from this population.

- **30%** of the honeybees in the sample have stripes and **10%** of the hornets in the sample have stripes.
- From previous research, you know that **80%** of the entire population are honeybees and the remaining **20%** are hornets.
- **65%** of the sample of bugs are honeybees and **35%** of the sample are hornets.

What is the **post-stratification estimate** of the **proportion** of all bugs in the population with stripes? You should assume that the assumptions of post-stratification are satisfied.

You do not need to simplify your answer, but if your answer is not valid Python syntax, you may lose credit.

For example: `(3)*(10)` and `0.25` are valid Python expressions. `(3)(10)` and `25%` are not valid Python expressions.

Post-strat estimate

e.g. `(3 + 2 - 1) * (4 / 5)`



`0.8 * 0.3 + 0.2 * 0.1`

Summer 2026 Past Quizterm 1 Questions

Question 1

In the first four lectures, we saw that **higher** rates of free/reduced price lunch at California public high schools were associated with (select all that apply):

- ☒ Higher rates of application to UC Berkeley in 2025
- ☐ Lower rates of application to UC Berkeley in 2025
- ☒ Lower rates of attending/enrolling at UC Berkeley in 2025
- ☐ Higher rates of attending/enrolling at UC Berkeley in 2025
- ☒ Higher rates of admission to UC Berkeley in 2025
- ☐ Lower rates of admission to UC Berkeley in 2025

Question 2

Consider a transactions dataset with the following columns:

Column	Description
<code>transaction_id</code>	a unique identifier for each transaction
<code>customer_id</code>	an identifier for the customer associated with the transaction
<code>store_id</code>	an identifier for the store where the transaction occurred
<code>date</code>	the date of the transaction
<code>year</code>	the year of the transaction
<code>total</code>	the total dollar amount of the transaction

Each row of the dataset represents one transaction.

Note: Customers can make more than one transaction at more than one store on a particular day.

Write an analysis plan for each of the questions below.

- Your analysis plan should start with the dataset above and result in a `DataFrame` that answers the question.
- Your final `DataFrame` must only include rows that are relevant to the question. No extra rows.
- However, your final `DataFrame` may optionally include columns that are not required to answer the question. For example, if you are asked to identify customers, but your answer contains a transaction ID column that is not used, that's fine.
- Your final `DataFrame` should require no additional computation to fully answer the question. A non-technical person should be able to answer the question by scrolling through your final dataframe.

Make sure to use a subset of the appropriate canonical English terms we learned in lecture. See reference sheet for examples.

(a) How many customers spent more than \$200 in the year 2024? [2 Pts]

B	I	U	G	”	“	x ₂	x ²	f _x	≡	≡	≡	≡	Normal	÷	Normal	÷	A	⌂	I _x
Your answer here																			
Filter to year == 2024. Group by customer_id and sum(total). Filter sum > 200. Count number of rows.																			

(b) What was the second largest transaction on each date? You can assume each date had at least 10 transactions. [2 Pts]

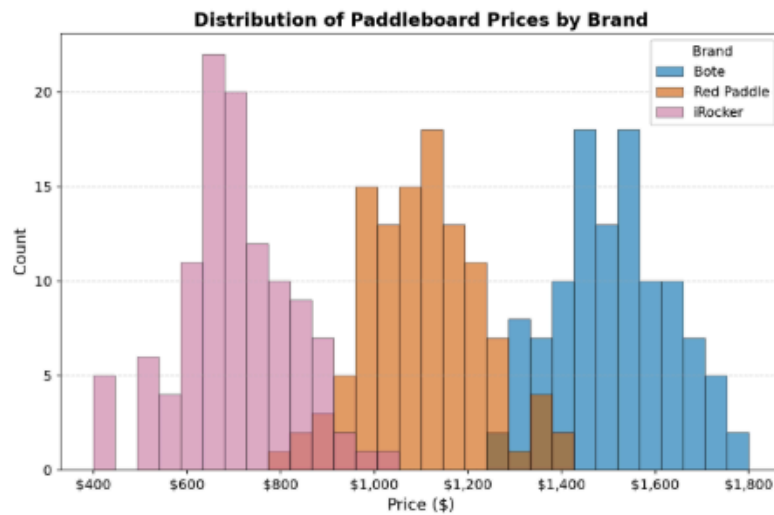
B	I	U	G	”	“	x ₂	x ²	f _x	≡	≡	≡	≡	Normal	÷	Normal	÷	A	⌂	I _x
Your answer here																			
Group by date and sort transactions within each date by total descending, take the 2 nd row per date.																			

(c) Which customer spent the most amount of money at each store? You can assume that, at each store, there is a single unique customer that spends the most amount of money. [2 Pts]

B	I	U	G	”	“	x ₂	x ²	f _x	≡	≡	≡	≡	Normal	÷	Normal	÷	A	⌂	I _x
Your answer here																			
Group by store_id and customer_id and sum(total). Within each store, sort by descending total and get the row with the max summed total within each store_id.																			

Question 3

The plot below shows **three overlaid histograms** of paddleboard sales.



Question

[2 Pts] What is the **minimum number of columns** `paddleboards` needs to have in order to generate this plot with a tidy `DataFrame`?

Your Answer

price, brand

integer 2

?

[2.5 Pts] Which of the following plot types is an appropriate choice for visualizing the same column(s) of data used by the three overlaid histograms? Select **True** if the plot can visualize the same data, and **False** if not. Assume that you cannot make any transformations or aggregations of the underlying data and you cannot use additional columns or data sources.

Visualization

Side by side boxplots

Overlaid KDEs

Contour Plot

Hex Plot

Lineplot

True / False

☒ True ☐ False [Clear](#)

☒ True ☐ False [Clear](#)

☐ True ☒ False [Clear](#)

☐ True ☒ False [Clear](#)

☐ True ☒ False [Clear](#)

Question 4

A streaming service's music catalog contains songs from only two genres: **reggae** and **jazz**. You pull a small sample of songs from this catalog.

- **15%** of the reggae songs in the sample are a hit and **5%** of the jazz songs in the sample are a hit.
- **20%** of your sample are reggae songs and **80%** of your sample are jazz songs.
- From the catalog metadata, you know that **40%** of the entire catalog is reggae and the remaining **60%** is jazz.

What is the **post-stratification estimate** of the **proportion** of all songs in the catalog that are a hit? You should assume that the assumptions of post-stratification are satisfied.

You do not need to simplify your answer, but if your answer is not valid Python syntax, you may lose credit.

For example: `(3)*(10)` and `0.25` are valid Python expressions. `(3)(10)` and `25%` are not valid Python expressions. Invalid syntax will be flagged when you save so you can fix it before your final submission.

Post-strat estimate e.g. `(3 + 2 - 1) * (4 / 5)`



$$(0.4) * (0.15) + (0.6) * (0.05)$$

Question 5

A fruit stand has a basket of 8 pieces of fruit, each labeled with a single symbol (@ = apple, X = dragonfruit, A = avocado):

@A@@X@AX

For each sample below (written as a length-3 or length-4 sequence of symbols), select **Possible to observe** if it's a possible sample according to the instructions above it, or **Impossible to observe** if it's not.

a) [0.25 Pts] A simple random sample (SRS) of size 3

- XXX ☐ Possible to observe ☒ Impossible to observe [Clear](#)
- @@@ ☒ Possible to observe ☐ Impossible to observe [Clear](#)
- X@A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

b) [0.25 Pts] Sample 3 pieces of fruit uniformly at random from the basket *with replacement*

- XXX ☒ Possible to observe ☐ Impossible to observe [Clear](#)
- @@@ ☒ Possible to observe ☐ Impossible to observe [Clear](#)
- X@A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

c) [0.25 Pts] A stratified random sample of size 4, stratified by fruit type.

There are three different fruit types in the basket.

- @@@@ ☐ Possible to observe ☒ Impossible to observe [Clear](#)
- @@XA ☒ Possible to observe ☐ Impossible to observe [Clear](#)
- @XXA ☒ Possible to observe ☐ Impossible to observe [Clear](#)

d) [0.25 Pts] Sample 3 pieces of fruit uniformly at random from the basket *without replacement*

- XXX ☐ Possible to observe ☒ Impossible to observe [Clear](#)
- @@@ ☒ Possible to observe ☐ Impossible to observe [Clear](#)
- X@A ☒ Possible to observe ☐ Impossible to observe [Clear](#)

Question 6

Aadil wants to determine whether more humid conditions are associated with different match results at the 2026 World Cup. For many matches, he records the `humidity` (%) during the match and the `result` for the home team (Win, Draw, or Loss).

Here are the first five rows of the data Aadil collected. There are 1000 rows in the full `DataFrame`:

	humidity	result
0	72.4	Draw
1	80.8	Loss
2	75.1	Win
3	72.2	Loss
4	66.2	Win

(a) Aadil wants to visualize the distribution of `humidity`. Mark **True** if the plot type is appropriate for this visualization task, and **False** otherwise. [0.5 Pts]

Histogram ☒ True ☐ False [Clear](#)

Violin plot ☒ True ☐ False [Clear](#)

KDE plot ☒ True ☐ False [Clear](#)

Bar plot ☐ True ☒ False [Clear](#)

(b) Aadil wants to visualize the relationship between `humidity` and `result`. Mark **True** if the plot type is appropriate for this visualization task, and **False** otherwise. [0.5 Pts]

Side-by-side boxplots ☒ True ☐ False [Clear](#)

Overlaid histograms ☒ True ☐ False [Clear](#)

Contour plot ☐ True ☒ False [Clear](#)

Hexplot ☐ True ☒ False [Clear](#)

(c) Aadil wants to visualize the distribution of `result`. Mark **True** if the plot type is appropriate for this visualization task, and **False** otherwise. [0.5 Pts]

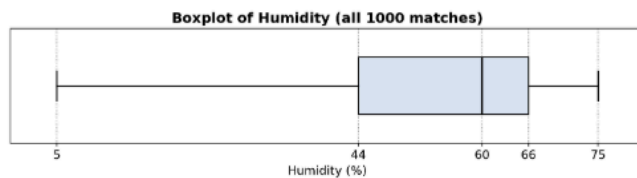
Bar plot ☒ True ☐ False [Clear](#)

Histogram ☐ True ☒ False [Clear](#)

Violin plot ☐ True ☒ False [Clear](#)

KDE plot ☐ True ☒ False [Clear](#)

(d) Below is a boxplot of `humidity` across all 1000 matches. Which of the following can be determined from this boxplot? **Select all that apply.** [1 Pt]



- ☒ The data is left skewed
- ☐ The data is right skewed
- ☐ The data is unimodal
- ☐ The data is bimodal
- ☒ The median of the data
- ☐ The mean of the data
- ☐ Approximately 75% of all humidity values in the dataset are above 66
- ☐ The 10th percentile of the data is 5

Question 7

There are simply too many episodes of **Love Island**, and Sarah does not have time to watch every recoupling. Instead, Sarah wants to use regular expressions to parse the transcripts.

Consider the following transcript, **which is written as a single long string**:

“gigi: Even though we've only known each other for 74 weeks, he gives me #loyalty, so the boy I want to couple up with is owen.

TK: Even though we've only known each other for 2,553 days, he gives me #energy, so the boy I want to couple up with is caleb.

Trinity: Even though we've only known each other for 4 weeks, he gives me #comfort, so the boy I want to couple up with is mason.”

For each regular expression in the table below, write the matching substring(s) from the transcript. Write matches in the order they appear in the text.

If there are more than 3 matches, write only the first 3. If there are fewer than 3 matches, write NA in the remaining boxes.

Represent any whitespace character in a match using \s (for example, a single space is written as \s).

Regex	Match 1	Match 2	Match 3
start ^\w+	gigi ?	NA ?	NA ?
\d{1,2}\d?	74 ?	2 ?	553 ?
[A-Z][^A-Z:\s]*:	Even ?	T ?	K ?